

The effects of computer-based cognitive training on attention and working memory in Thai older adults with mild cognitive impairment

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ABSTRACT

Background: Mild cognitive impairment (MCI) is an intermediate state occurring between healthy aging and dementia among the elderly which result in cognitive, especially regarding attention and working memory. However, the declined state can be abated if training or stimulations are promptly promoted by a computer-based cognitive training program.

Objective: The aim of this study was to explore the effects of the Neofect Cognition as a computer-based cognitive training program on attention and working memory in Thai older adults with MCI.

Materials and methods: Twenty-three Thai older adults with MCI were divided into two groups; the experimental group (N=13) receiving the Neofect Cognition program for 6 executive weeks, 3 times per week, 45 minutes per session, and the control group (N=10) receiving education about MCI. Data was collected by assessing before and after participating in the program using the Trail Making Test (TMT) and the Memory subtest of the Thai Cognitive-Perceptual Test (Thai-CPT).

Results: The results showed that the experimental group had a statistically significant difference of attention (TMT-B) and working memory scores, while the control group had no statistically significant difference for both attention and working memory. Furthermore, there was a significant difference between the groups in attention (TMT-B) and working memory.

Conclusion: From the results, it can be stated that the computer-based training program (Neofect Cognition) might improve attention and working memory in Thai older adults with MCI.

Introduction

Mild cognitive impairment (MCI) is a condition in which individuals show cognitive impairment with minimal impairment of daily living, is in an intermediate state between healthy aging and dementia.¹ MCI is considered to be a 6-30 % risk for developing dementia.² Clearly, MCI is a risk factor for dementia.³⁻⁵ A review of the prevalence of MCI worldwide was published in 2020-2023, which showed it arising at a rate of 12 to 71.4 % and that high-risk factors associated with MCI were age and educational level.⁶⁻⁹ MCI causes cognitive deterioration and distress in family, caregivers, and the economy of a country.¹⁰ However, the declined state can be abated if training or stimulations known as "cognitive interventions" are promptly promoted and utilized.^{1,11}

Cognitive interventions aim at improving or maintaining cognitive abilities¹² and are divided into 2 types:

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pharmacological and nonpharmacological cognitive intervention.^{13,14} The pharmacological cognitive interventions have potential agents for slowing or reversing cognitive decline but have limitations in the efficacy of treatment, and evidence suggests that these do not improve cognitive function.¹⁴⁻¹⁶ This is the key reason for paying particular attention to nonpharmacological treatment.¹⁴ Previous studies have shown that older adults with MCI improved cognitive function after receiving nonpharmacological treatment.^{14,17-19} Examples of non-pharmacological interventions are physical exercise, music therapy, behavior therapy, education and group training, and technology-based training.

Nowadays, technology-based training is widely used for improving cognitive function. It is not only useful for improving cognitive function but also has several advantages over traditional intervention.^{15,18,20-23} For example, it can motivate patients and is enjoyable; it is flexible and can be personalized; and it provides immediate feedback using a game format. Moreover, the computer-based training also has positive effects on psychosocial function in older adults with MCI.²¹ Based on the benefits of the computer-based training program, the rapid growth in the number of older adults in Thailand,²⁴ and the problems resulting from MCI, we employed a computer-based cognitive training program having Thai cultural aspects that can be tailored to target specific cognitive domains affected by MCI.²⁵

The Neofect Cognition is a computer-based game used for improving cognitive functions.²⁵ It can be used to stimulate 4 domains of cognition, including attention, memory, executive function, and visuospatial. The Neofect Cognition consist of 10 games in each domain. For example, 1) catching chameleon; game in attention domain; challenges patients to pay selective attention and divided attention, 2) making phone call; game in memory domain; challenges patients to pay working memory, 3) catching mole; game in executive function domain; challenges patients to use organizational skills and problem solving and 4) separate image; game in visuospatial domain. Previous studies in Asian countries involving various groups of participants showed that the Neofect Cognition has beneficial effects on cognition for stroke patients,²⁶ brain injury patients,²⁷ Alzheimer's patients,²⁸ ADHD children,²⁹ and high-functioning autism spectrum disorder.³⁰ Nevertheless, applying computer-based cognitive training with older adults with MCI in Thailand has been limited, especially in occupational therapy profession. Although the Neofect Cognition was developed in Asia, there is a lack of evidence, which has investigated the effectiveness and feasibility of the Neofect Cognition for Thai older adults with MCI.

The aim of this study was to investigate the effect of computer-based cognitive training using the Neofect Cognition on attention and working memory in Thai older adults with MCI.

Materials and methods

A quasi-experimental design, controlled study with pre-test and post-test was conducted with 23 Thai older adults having mild cognitive impairment (MCI) in Chiangmai, Thailand, from December 2021 to October 2023. This study was approved by the Ethics Committee of the Faculty of Associated Medical Sciences, Chiang Mai University (Study Code: AMSEC-64FB-006). The participants were divided into two groups with purposive sampling method: 1) the control group, which received education about MCI; 2) the experimental group, which received education about MCI and a computer-based training program using the Neofect Cognition for 6 executive weeks, 3 times per week, 45 minutes per session. The diagnosis used for MCI was based on the DSM-5, conducted by the geriatric psychiatrist through a clinical interview.

Inclusion and Exclusion criteria: This study included the participants as the following criteria: 1) age 60-79 years; 2) elementary education level 4 (Prathom 4 in Thai education system) or upper; 3) none of stroke and brain injury history; 4) capability with normal arm movement; 5) none of visual and auditory problems; 6) The Barthel index for activity of daily living scores equal to 12 or greater; 7) MSET-10 scores over 17; 8) MoCA-B scores below 25; 9) Thai version of the Geriatric Depression scores below 6; 10) no dementia; and 11) the consent obtained to the participation in computer-based intervention. Furthermore, the withdrawal criteria were obtained as the followings: 1) participants stating a possibility of quitting; 2) participants experiencing limited movement or brain injury while participating in the program. Additionally, the participants who were absent more than 3 times and/or those who currently used other cognitive training programs, would be excluded from the study.

The sample size estimation was derived from a related conducted by Kim and co-authors.³¹ The G*Power program (effect size=0.77, power of test=0.95, α =0.05). The calculation produced 21 people in each group, and we added 20% in case of dropout. Finally, 50 older adults with MCI were recruited. One hundred and forty people were invited to participate in the study; 101 were excluded due to unwillingness to participate, and 12 were excluded because they did not meet the criteria for the MCI. Finally, 27 participants were included in this study; 15 were in the experimental group and 12 in the control group. However, during the intervention program periods, 2 in experimental group were excluded due to unwillingness and absent to received program more than 3 times. Moreover 2 in control group were excluded due to accident. Finally, leaving 23 participants for the final analysis (Figure 1).

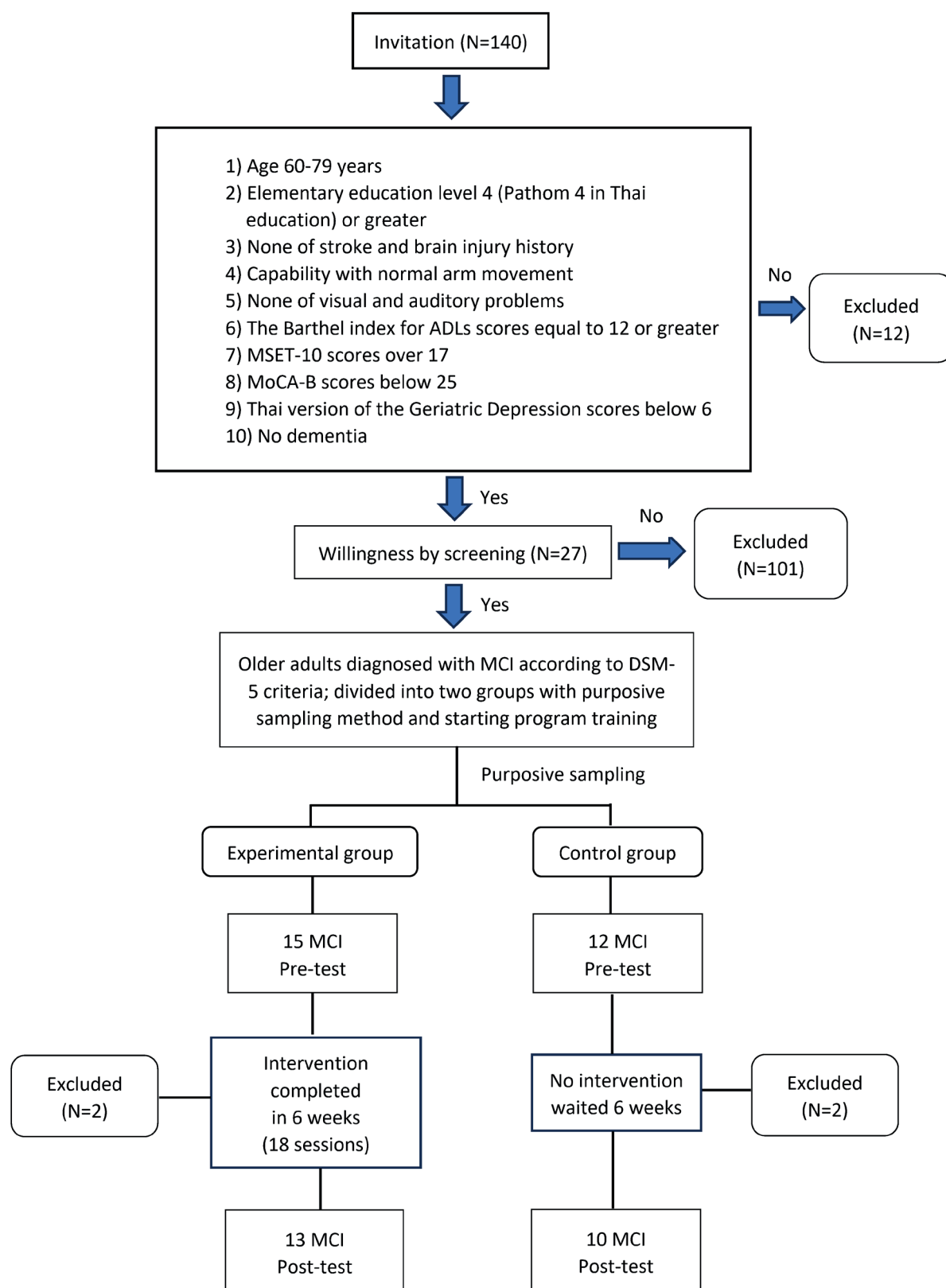


Figure 1. Flowchart of this study.

Measurements

Screening Tools

Barthel ADL index

Barthel ADL index is used to assessed physical performance in activities of daily living (ADL). Thai version of Barthel ADL index used in patients with low-energy femoral neck fracture has good interobserver and intra-observer reliabilities (0.71 and 0.96, respectively). Additionally, the internal consistency was acceptable (Cronbach's $\alpha=0.69$).³²

Mental status examination Thai 10 (MSET-10)

Mental Status Examination Thai 10 (MSET-10) is a screening test for dementia. It has excellent sensitivity and specificity. Additionally, it had been suggested that MSET-10 is appropriated for using in illiterate adults. It was adapted from Mini-Mental state exam-Thai 2002 (MMSE-Thai 2002) by Department of Psychiatry, Faculty of Medical of Chiang Mai University.³³

Montreal cognitive assessment-basic (MoCA-B)

Montreal Cognitive Assessment-Basic (MoCA-B) is a screening test for Mild Cognitive Impairment in Thai illiterate older adults.³⁴ It was developed as a collaboration among the MoCA Clinic and Institute in Canada, the Prince Mahidol Award Foundation, and Faculty of Medical of Chulalongkorn University in Thailand. MoCA-B evaluates 6 cognitive domains: visual perception, executive function, language, attention, memory, and orientation. It has good to excellent sensitivity, specificity, test-retest reliability, and internal consistency (0.81, 0.86, $r=0.91$, and $\alpha=0.82$, respectively)

Geriatric depression scale-15 (GDS-15)

Geriatric Depression Scale-15 (GDS-15) is a test of screening for depressive symptoms in older adults.³⁵ It has good reliability and validity. The Thai version of the Geriatric Depression Scale (TGDS) has good psychometric properties,³⁶ good internal consistency ($\alpha=0.85$).

Outcome measurement

The outcome was measured within 1 week before and after the intervention. The assessments were conducted by the trained therapists, 1 clinical psychologist and 1 occupational therapist, who blinded the group allocation.

Trail making test (TMT)

Trail Making Test (TMT), i.e., TMT-A and TMT-B, is a neuropsychological test of visual attention and task switching. It consists of 2 parts. In part A, it requires participants to connect the circles in a sequence as quickly as possible using a single continuous line. In part B, it requires participants to draw a line connecting several alternating letters in sequence. It has average to good test-retest reliability in parts A and B ($r=0.36-0.79$ and $0.44-0.89$, respectively).^{37,38} In this study it was used for assessing attention by a clinical psychologist.

The Thai cognitive-perceptual test (Thai-CPT)

The Thai Cognitive-Perceptual Test (Thai-CPT) is a perceptual-cognitive outcome measure for patients with brain injuries in Thailand. The Thai-CPT consists of 6 subtests which includes Visual Perception, Body Scheme Perception, Praxis, Memory, Matching and Categorization, and Problem Solving. It has good to excellent test-retest reliability (ICC=0.83-0.97), excellent inter-rater reliability (ICC=0.91-1.00), and acceptable internal consistency ($\alpha=0.78-0.96$).³⁹ This study used a Memory subtest for working memory with assessment by an occupational therapist.

The assessors were blinded from the participants in both groups.

Statistical analysis

Descriptive statistics included frequency and percentage. Cognitive outcomes in attention and working memory were assessed using the Wilcoxon signed-rank test to compare the effects of the intervention program between pre-test and post-test, the Mann-Whitney U-test to compare the effect of the intervention program on post-test in both groups, and the Chi-square to compare age and levels of educational between group.

Results

A total of 23 participants were included in this study, with 10 assigned to the control group and 13 to the experimental group. Participants in both groups were carefully matched based on key demographic variables including sex, age, and educational level, to ensure comparability. The demographic characteristics are summarized in Table 1.

Table 1. Characteristic of the participants in this study.

Variables		Control group (N=10)	Experimental group (N=13)	p value
Gender	Female	10 (100%)	11 (84.62%)	1.47
	Male	0 (0%)	2 (15.38%)	
Age range (years) Mean±SD 67.47±3.92	60-69	8 (80%)	9 (69.24%)	1.05
	70-79	2 (20%)	4 (30.76%)	
Status	Single	2 (20%)	2 (15.38%)	0.41
	Married	7 (70%)	9 (69.24%)	
	Divorce	1 (10%)	2 (15.38%)	
Levels of education	Level 4 (Prathom 4 in Thai education system)	10 (100%)	12 (92.30%)	1
	More than level 4	0 (0%)	1 (7.70%)	

The control group had a mean age of 66.50±3.84 years, while the experimental group had a mean age of 68.23±3.98 years. The Chi-square test indicated no significant difference in age between the two groups. In terms of educational background, all participants in the control group (100%) had attained an educational level of grade four, while in the experimental group, 12 participants (92.30%) had completed grade four, with one participant (7.70%) having a higher level of education. Overall, the demographic data revealed no statistically significant differences between the control and experimental groups across any of the measured variables, ensuring that both groups were comparable in terms of baseline cognitive ability before participating in the intervention.

The cognitive scores comparing the control and experimental groups after the intervention are presented in Table 2. The Mann-Whitney U test identified statistically significant differences between the groups in terms of performance on the TMT-B and working memory, with the experimental group demonstrating a higher mean rank compared to the control group in working memory and lower mean rank in TMT-B. And range of working memory scores of the control and experimental groups after the intervention are 20-32 and 26-32, respectively. These findings suggest that participants in the experimental group who underwent the Neofect Cognition intervention showed improved cognitive abilities, particularly in areas related to task-switching and working memory, as assessed by TMT-B and Thai-CPT.

Table 2. Post-intervention assessment between groups.

Test	Control group			Experimental group			Mann-Whitney U test	
	Mean (SD)	Mean rank	Sum of ranks	Mean (SD)	Mean rank	Sum of ranks	Z	p value
TMT-A*	-0.5 (33.88)	12.40	124	1 (26.57)	11.69	152	-0.314	0.39
TMT-B*	10.6 (39.41)	15.20	152	-11.92 (18.43)	9.54	124	-1.960	0.02
Thai-CPT**	-1.6 (1.50)	5.9	59	3.53 (3.30)	16.69	217	3.049	0.01

Note: *used for assessing attention, unit of performance is "seconds"; **used for assessing working memory, unit of performance is "score"; The scores range in Thai-CPT (memory subtest) is 0-34.

In contrast, no significant difference was observed between the groups in the TMT-A times, indicating that the intervention may not have had a notable effect on simpler attention tasks that primarily involve visual scanning and processing speed. The results imply that Neofect Cognition could be especially effective in enhancing alternating attention, as evidenced by the superior performance in TMT-B, as well as improving working memory among the participants in the experimental group. These findings provide promising evidence for the potential application of Neofect Cognition as a computer-based cognitive training tool aimed at improving specific cognitive functions in populations with similar needs.

Tables 3 and 4 present the statistical analysis and comparisons of cognitive scores within the control and experimental groups, respectively. In the control group, the comparison of pre- and post-test scores using the Wilcoxon Signed-Rank test revealed no statistically significant differences in TMT-A, TMT-B, and working memory, as

presented in Table 3. This suggests that the control group did not exhibit notable cognitive improvements across these measures over the course of the study. Conversely, in the experimental group, significant differences were observed in TMT-B and working memory scores, indicating that participants in this group experienced meaningful cognitive enhancements in these areas. However, TMT-A times in the experimental group did not show a statistically significant difference (Table 4).

When examining the mean rank scores, the experimental group consistently outperformed the control group across all cognitive measures. These results suggest that the experimental group demonstrated improvements in both attention and working memory following participation in the intervention program. The findings imply that the Neofect Cognition program may be particularly effective in enhancing alternating attention, as required by the TMT-B, and in improving working memory.

Table 3. Pre- and post-intervention assessment of control group (N=10).

Test	Control group (N=10)					
	Pre-test	Post-test	Wilcoxon Signed-Rank Test			
	Mean (SD)	Mean (SD)	Mean rank	Sum of ranks	Z	p value
TMT-A	121 (53.91)	120.5 (54.35)	4.67	28	-0.015	0.50
TMT-B	117.7 (28.37)	128.3 (52.27)	4.30	21.5	-0.612	0.74
Thai-CPT	28 (3.09)	26.4 (3.50)	0	0	-2.55	0.99

Table 4. Pre- and post-intervention assessment of experimental group (N=13).

Test	Experimental group (N=13)					
	Pre-test	Post-test	Wilcoxon Signed-Rank test			
	Mean (SD)	Mean (SD)	Mean rank	Sum of ranks	Z	p value
TMT-A	79.08 (29.04)	80.07 (24.79)	5.38	43	-0.314	0.39
TMT-B	104.69 (33.84)	92.76 (25.65)	8.17	73.5	-1.960	0.02
Thai-CPT	25.38 (3.70)	28.92 (1.97)	7.42	89	3.049	0.001

Discussion

The results demonstrated that the Neofect Cognition might help older adults who are suffering from MCI with attention and working memory impairment. After the experiment, the experimental group showed that the Neofect Cognition provided more positive results than in the control group on attention and working memory both within (Tables 3 and 4) and between groups (Table 2). These results were obtained after assessment using the TMT-B and the Thai-CPT. The results suggest that computer-based cognitive training using the Neofect Cognition may be effective for improving attention and working memory

in the experimental group. This confirms previous studies that support the beneficial effects of computer-based cognitive training on attention and working memory.^{12,40-43} It is likely that the Neofect Cognition is preferable due to its ability to specifically and reliably address attention and working memory. In addition, the games in the Neofect Cognition were assessed for content validity in Thai context by five experts who determined its suitability before the experimental group used these games.

Moreover, despite the small sample size, these results are like the results of previous studies, indicating that the computer-based cognitive training program affect cognitive

ability. Based on the results of previous studies of the characteristics of Alzheimer's dementia patients conducted by Hwang and a co-researcher,²⁸ it was shown that the computer-based cognitive training is effective in improving memories of Alzheimer's dementia patients. Kim and co-researcher,³¹ and Finn and McDonald⁴⁰ reported that computer-based cognitive training program increased cognitive ability, as in our study.

However, from the results of the TMT-A portion, there was no significant difference within groups (Tables 3 and 4) or between groups (Table 2). Likewise, Yang and a co-researcher found that attention times in the experimental group did not improve after experimentation.¹⁰ It is reasonable to assume that insufficient attentive intervention in TMT-A is related to inferior visual scanning and sustained attention compared to TMT-B's alternate attention and divided attention formats. To clarify, in cases where the experimental group had received the attention training based on a computer-based cognitive training program for a longer period, the attention times would likely decrease.

Additionally, when considering the baseline attention of both groups before the experiment, the cognitive function of the TMT-A was significantly different before the experiment ($p=0.02$) and would notably influence the attention times in the only slightly different times obtained between groups. Moreover, when we consider the results of the TMT-B and the TMT-A, it was found that differences had a large mean and standard deviation. This could most likely be the consequence of the purposive sampling method being chosen, which led to the differentiation among the samplings in terms of attention, and of failure to abort specific criteria for sampling participants. In the meantime, the control group having not received the computer-based cognitive training, was prone to be derive a declination in the cognitive ability as the result of pathological effects of disease; in the other word, MCI is a cognitive deterioration.^{1,44,45} Among the participants, as a matter of fact, there are personal diseases such as hypertension and diabetes mellitus confounding factored to the cognitive ability.^{5,41}

This study had some limitations due to the small sample size which may have affected the ability to examine the absolute effects of computer-based cognitive training on MCI. Furthermore, due to restricted conditions along with little variation in educational levels, the results could not be generalized for other populations. Lastly, since the data was collected in a specific area, being a community in Chiangmai, there are clear limitations. For the further studies, it is recommended the data be gathered in more diverse populations residing in other areas. Moreover, using higher measurement to accurately represent the brain area responding to each domain of cognition while using the Neofect Cognition would be benefit for the further study.

Conclusion

The result from this study showed that the Neofect Cognition might help improve attention and working memory in older adults with MCI. These findings confirmed the

effectiveness of computer-based cognitive training which could be helpful for older adults with MCI to improve attention and working memory and might prevent the onset of dementia when stimulating activities are regularly and sequentially provided.

Conflict of interest

The authors have no conflicts of interest to declare.

Ethical approval

This study was approved by the Ethics Committee of the Faculty of Associated Medical Sciences (study code AMSEC-64FB-006; approval date July 8, 2021).

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